

Design Report

by

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Executive Summary

Purpose

The aim of this project was to design a compact drivetrain for the Formula 24 (F24) Kit Car that enables a top speed of 20–24 km/h, fits within the restricted rear chassis space, and remains compatible with the vehicle's quick-release mounting system. The new drivetrain was required to be reliable, manufacturable within the Student Training Workshop, and easy to assemble and maintain.

Approach

The project followed a structured 10-week design process involving concept generation, engineering analysis, detailed component selection, procurement, CAD development, manufacturing, assembling and performance verification. Multiple drivetrain concepts were evaluated using weighted criteria for safety, manufacturability, durability, logistics, and cost, leading to the selection of a three-stage system using two simplex 06B chain stages and a spur-gear stage. Speed and torque analysis determined the required modification ratio of 10.12, while AGMA gear stresses, shaft bending and shear stresses, and bearing dynamic load ratings were calculated to ensure mechanical reliability. Iterative CAD modelling resolved spatial constraints, defined shaft layouts, and finalised tensioning and mounting methods. Components were procured from RS, and the prototype is manufactured and assembled using STW facilities. Bench tests were used to assess manufacturing quality, and an on-track test will take place later in June 2026 to verify chain stability, gear meshing, thermal behaviour, and the final output speed, confirming that the drivetrain meets the design requirements.

Final Design

The selected solution is a three-stage drivetrain providing a modification of 10.12. It consists of:

- Stage 1: 12T → 30T sprockets
- Stage 2: 18T → 60T spur gears
- Stage 3: 14T → 17T sprockets

Two custom shafts support the gears and sprockets, mounted in SKF P20 pillow block bearings attached to an aluminium extrusion cassette as shown in Figure A.

The gears/sprockets selected have a safety factor of at least 2; shafts have a safety factor of 5.8 under operational loading. Keys are used to transfer torque; shoulders, circlips and endcaps are used to axially constrain the gears and

an idler tensioner ensures stable chain engagement in Stage 1. The drivetrain fits fully within the F24 rear chassis and remains compatible with the quick-release module.

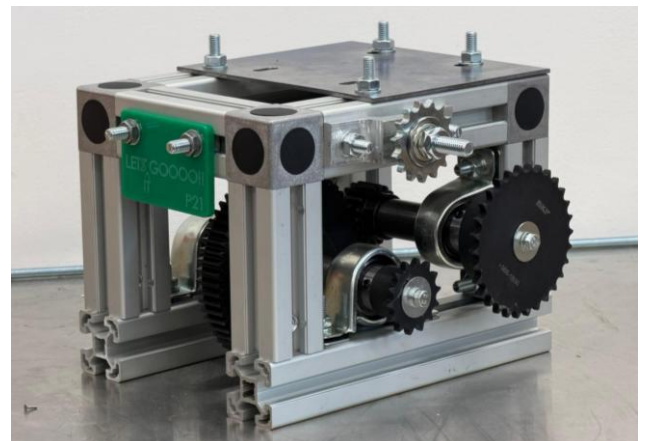


Figure A: Transmission module prototype.

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1. Introduction

The Formula 24 (F24) Kit Car is a single-seat, electric-powered vehicle developed by the Greenpower Education Trust to promote STEM engagement and inspire future mechanical engineers. The standard F24 drivetrain uses a single-stage chain and sprocket transmission, directly connecting the electric motor to the rear axle.

The objective for this project is to provide the client an alternative drivetrain module capable of maintaining the car's top straight-line speed at 20–24 km/h, while ensuring reliability and ease of maintenance. The proposed design employs a three-stage transmission with an overall modification of 10.1, using a combination of gears and sprockets. The entire assembly is designed to fit within the rear chassis of the F24 Kit Car.

This project was carried out by a team of five second-year Mechanical Engineering students at Imperial College London. Thomas Sutherland served as the Project Manager, ensuring the project is on track. Jayden Lee acted as the Student Training Workshop (STW) Liaison, holding communication regarding manufacturing constraints and processes. Ernest Collinson was the CAD Lead, responsible for reviewing and scrutineering CAD models and engineering drawings. Arany Nanthakumaran served as Documentation Manager, overseeing the management of documents, spreadsheets, and meeting notes. Charles Hung acted as Tutor Liaison, having discussions with the tutor regarding project details and design decisions to ensure design decisions meet project requirements. All team members contributed to calculations, CAD development, engineering drawings, and documentation.

The report below presents the development of the drivetrain solution, including conceptual design, component selection and justification, product specifications, embodiment design, manufacturing and prototyping, risk assessment, and project management.

2. Product Design Specification

This Product Design Specification (PDS) outlines the requirements and constraints for the drivetrain transmission module developed for the F24 kit car (Table 1). It defines the expected operational behaviour, manufacturing limitations, safety compliance, and verification methods.

Table 1: PDS for the drivetrain module.

Version	Details of Change	Date
1.0	Initial release.	13/10/2025
2.0	Verification methods clarified.	18/11/2025

Element	Criteria	Verification
Customer		
Needs	The transmission module must be designed to be easily attached and released from the car.	Dry Testing
Competition	The drivetrain fulfils target operations on F24 kit car.	Testing
Operation		
Performance	Achieve a top speed of 20-24 km/h on straight roads. Must be travelling at < 25 km/h at any given moment.	Straight-line Testing
Environment	Be able to handle vibrations due to road surface. Operational temperature ranges from -5°C to 40°C.	Testing
Size	Must fit within the space of the back frame of the F24 kit car.	Design Review
Life		
Service Life	Reach the same service life of the motor around 30,000 hours.	Design Review
Maintenance	Lubrication of moving parts should be done before initial drive and subsequent uses.	Testing
Producer		
Product Cost	Total cost of all purchased parts must be less than £300.	Design Review
Manufacturing Constraints	At least one part must be manually turned. At least one part must be manually milled. At least one part must be laser cut. A maximum of three parts can be 3D printed. Only one part can be manufactured using the CNC. Parts can only be purchased from RS Components.	Design Review
Regulatory		
Safety Standards and Regulations	Complies with safety standards from the Imperial College London Department of Mechanical Engineering, the Health and Safety Executive (HSE), and relevant ISO machinery safety standards. All components must be securely fastened, and no moving transmission elements may be exposed.	Design Review and Inspection
End of Life	Reuse and recycle all parts. Useful parts should be donated to Department of Mechanical Engineering and Workshop.	Design Review

3. Concept Design

3.1 Concept Generation

The proposed drivetrain must fit within the fixed rear frame of the F24 Kit Car and deliver a speed ratio that enables the vehicle to reach 20–24 km/h. To retain compatibility with the existing quick-release system, the new drivetrain must be mounted to a cassette assembled from aluminium extrusions provided by the client. The current drivetrain uses a 12-tooth 06B-1 sprocket on the motor and a 72-tooth sprocket on the axle, together with an additional 17-tooth sprocket that was later supplied by the client.

The F24 car has a dry mass of 113 kg, and its total operational mass is estimated to be 183 kg including the driver and drivetrain. Using a rolling resistance coefficient of 2%, the resulting frictional force is 35.90 N. A conservative air-drag estimate of 4.72 N is also applied. Given the tyre radius of 0.254 m, the required output torque at the axle is calculated to be 10.32 Nm.

The vehicle is powered by a 24 V DC motor that receives full voltage immediately upon activation, as the system has no-speed controller or soft-start capability. Motor performance characteristics used in the design process are provided in Appendix G. Based on the motor data and assumed efficiency of 93%, the input and output speeds, torques, and the ideal overall transmission ratio are calculated in Appendix G and summarised in Table 2. With these values, two concepts are created and discussed.

Table 2: The input and output speed, torque, power, and ideal modification.

	Speed [rpm]	Torque [Nm]	Basic Power [W]	Required Power [W]	Ideal Modification
Input	2076.5	1.12		242.3	9.94
Output	208.9	10.32	225.7		

3.1.1 Concept 1: Two Stage Design

The first design incorporates only a two-stage sprocket and chain transmission as shown in Figure 1. This solution is simple, cost-effective, and light weight. Additionally, the chains are flexible for positioning the motor. Quick replacement of the chains is possible, and it could handle the high torque shock from the motor start up.

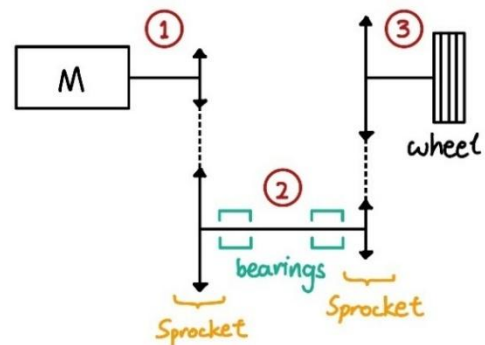


Figure 1: Schematic drawing of the two-stage concept.

3.1.2 Concept 2: Three Stage Design

The second design uses a three-stage transmission with a combination of sprockets and spur gears as shown in Figure 2. This configuration adds more cost and weight, but the spur gears can handle higher torque. This concept is ultimately selected. The reasons will be discussed in Section 3.2.

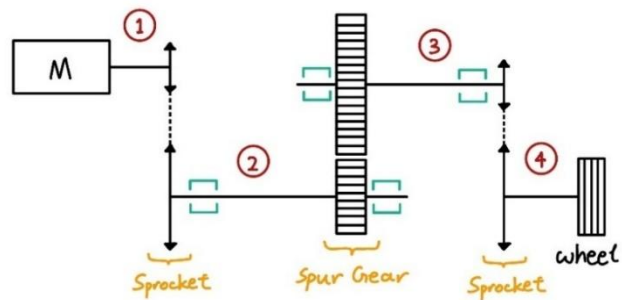


Figure 2: Schematic drawing of the three-stage concept.

3.2 Concept Comparison and Selection

To identify the most suitable drivetrain concept, each was evaluated against weighted criteria from the project PDS as shown in Table 3. The criteria targeted factors critical to the drivetrain's feasibility, reliability, and manufacturability. The two-stage concept scored highest overall due to its simplicity, ease of manufacture and maintenance, and would have been selected. However, the client's requirement for a fixed 17-tooth output sprocket would force an excessively large first-stage sprocket, leading to chain-wrap issues and limited component availability. Therefore, the three-stage concept was chosen as the final design, offering greater ratio flexibility and access to suitable off-the-shelf gears and sprockets.

Table 3: Selection criteria showing the overall score for both concepts.

Selection Criteria	Weight	Two-Stage		Three-Stage	
		Rating	Score	Rating	Score
Safety	0.20	4	0.8	5	1
Ease of manufacture	0.20	5	1	4	0.8
Ease of maintenance	0.05	5	0.25	4	0.2
Durability	0.15	3	0.45	4	0.6
Tolerance to error	0.15	4	0.6	3	0.45
Part availability	0.20	5	1	5	1
Cost	0.05	4	0.2	3	0.15
Total Score		4.3		4.2	
Rank		1		2	

4. Mechanical Design

4.1 Embodiment Design

As specified in the PDS, the drivetrain and motor must fit within the rear chassis of the F24 car. The assembly is supported by aluminium extrusions. With the three-stage design selected, the overall reduction is distributed across the motor sprockets, spur gears, and axle sprockets. Two shafts carry the gears, sprockets, and bearings that mount to the extrusion frame. In the following pages, the final design will be presented and justified. Design iterations will be discussed in Section 4.2.

4.1.1 Stage Ratio, Gears, Sprockets, and Chains

The details of each stage's gears and sprockets is shown in Table 4.

Table 4: Details on modification, gears, and sprockets for each stage.

Stages	Modification	Spur Gear Module 2		Sprocket 06B-1		Centre Distance [mm]	Chain Length [Pitches]
		Input	Output	Input	Output		
1	2.50			12T	30T	158.27	56
2	3.33	18T	60T			78	
3	1.21			14T	17T	268.95	72
Overall	10.12						
Material		Steel EN8 080M40/080M46					

Non-integer modification coefficients were applied to each stage to promote even wear. Gear strength was assessed by calculating the minimum face width using the Lewis equation in Equation 1, where F is the transmitted load, K_v the dynamic factor, m the module, Y the Lewis factor, and σ_p the permissible

material stress (Ihracska et al., 2023). This yields the tooth-bending safety factor. Contact stress σ_c and its safety factor S_H were then obtained using Equations 2 and 3 (AGMA, 2004). Full calculations are provided in Appendix H, and the resulting safety factors are summarised in Table 5.

$$w = \frac{F}{K_v * m * Y * \sigma_p} \quad (1), \quad \sigma_c = C_p \sqrt{F K_0 K_V K_S \frac{K_m C_f}{w d I}} \quad (2), \quad S_H = \frac{S_C Z_n C_H}{\sigma_c K_T K_R} \quad (3)$$

Table 5: Safety factor for both gears due to tooth bending and contact stress.

	Safety Factor	
	18T Gear	60T Gear
Tooth Bending	8.16	11.43
Contact Stress	2.12	2.27

The 12T and 17T sprockets currently on the vehicle uses simplex 06B. To make sure the sprocket stages are strong enough, the application factor f_1 , tooth factor f_2 , and design power P_D is found to guarantee that they are sufficient to carry the load.

The chain length in pitches L_p is calculated using Equation 4 where z_1 and z_2 are number of teeth, CD is the centre distance between the sprockets, and p is the pitch.

$$L_p = \frac{z_1 + z_2}{2} + \frac{2CD}{p} + \frac{\left(\frac{z_2 - z_1}{2\pi}\right)^2 * p}{CD} \quad (4)$$

4.1.2 Bearings

SKF P20 pillow block bearings with a 20 mm internal diameter are used. The drivetrain requires a 30,000-hour service life, and each bearing has a dynamic load rating C of 12700 N. Radial loads on bearings A–D were calculated, and the minimum required rating C_{min} was determined using Equation 5, based on the load F , life L_{10} , and bearing constant k (Mohammed, 2024). As shown in Table 6, all bearings operate below their rated capacity. Full calculations are in Appendix I.

$$C_{min} \geq F(L_{10})^{\frac{1}{k}} \quad (5)$$

Table 6: Key properties on shaft and bearing load.

Shaft	Revolutions Per Minute [$rad\ s^{-1}$]	Bearings	Equivalent Radial Load [N]	Minimum Required Dynamic Load [N]
1	832.4	A (floating)	124.67	1427.12
1	832.4	B	140.62	1609.70
2	249.7	C (floating)	110.14	843.38
2	249.7	D	788.73	6039.63

4.1.3 Shafts

Both shafts are made from steel EN1A to handle the high torque and are nearly identical to ease manufacturing. Shaft dimensions are shown in Table 7 with the general diagram.

The shafts include keyways, circlips, and a threaded hole. Further details on fasteners are provided in Section 4.1.5. Shaft 1 is slightly longer than Shaft 2 so that the sprockets sit on different axial planes.

Table 7: Shaft 1 and 2 diameters (D) and lengths (L) for each section.

*All dimensions in millimetres

Shaft 1				Shaft 2			
D1	20	L1	110	D1	20	L1	110
D2	24	L2	42	D2	24	L2	42
D3	20	L3	71.45	D3	20	L3	54

Stress analysis was performed on Shaft 2 as it has the heaviest loads. Maximum bending and shear stresses were calculated. Table 8 summarises the key results, showing that the shaft designed provides sufficient strength. The calculation process is given in Appendix J.

Table 8: The key stress values and safety factor for the Shaft 2.

	Maximum bending stress σ_b [MPa]	Maximum shear stress τ [MPa]	Equivalent von Mises stress σ_{vm} [MPa]	Allowable stress of material [MPa]	Safety factor
Shaft 2	38.2	8	39.8	230 (Smiths Metal, 2023)	5.8

4.1.4 Shaft and Motor Layout

Throughout the design process, the shaft and motor positions were revised several times. In the final layout, the motor is mounted on top of the extrusion cassette, which can be seen in Figure 3. It is the only viable location, as its large profile would otherwise interfere with the safety cover. Mounting it in the original F24 position was also ruled out due to chain interference with the car frame geometry.

The first shaft and its subassembly are mounted on the vertical extrusions closest to the motor, while the second shaft is mounted on the lower extrusions, as shown in Section 4.3 Figure 3.

4.1.5 Keys, Circlips, Fasteners, and Tensioner

Steel keys are used to transmit torque to the gears and sprockets. The gears and the 30T sprocket use 6×6×25 mm keys, while the 14T sprocket uses a 6×6×18 mm key. A 6 mm width was selected based on standard recommendations for 20 mm shafts.

Both gears are positioned using a shoulder and an S020M circlip. Circlips were chosen because the gears sit near the shaft centre, making other retention methods impractical, and axial loads are minimal. The bearing and sprocket at the short end of each shaft are secured with an endcap and an M8×25 mm screw, with plain and spring washers to prevent loosening. An aluminium spacer is placed between the bearing and the 30T sprocket to accommodate the extended shaft.

Components mount to the aluminium extrusions using M8 T-slot nuts and M8×16 mm socket screws. The motor attaches to the mounting plate using M8×30 mm and M8×25 mm screws.

A tensioner subassembly adjusts the first-stage chain, consisting of an idler sprocket on an M10×60 mm bolt, locked with thin hex nuts and supported by an aluminium bracket.

4.2 Design Iterations

Iterations to stage ratios and its reasons is shown in Table 9.

Table 9: Iterations over teeth count for each stage.

Iteration	Stage 1 (Sprockets)	Stage 2 (Spur Gears)	Stage 3 (Sprockets)	Reason
1	12-35	18-60	17-17	Initial design giving overall ratio of 10; stages balanced and similar.
3	12-36	18-50	14-17	35T sprocket unavailable (out of stock). 14T prevents sprocket interference.
4	12-30	18-60	14-17	Shaft layout rearranged to fit 60T gear. Smaller 30T sprocket increases chain wrap

Both shafts were designed with identical step dimensions, except for the outermost step length, enabling faster CNC machining and preventing sprocket interference. After finalising stage ratios, shaft dimensions were updated for proper bearing fit. The initial use of 22 mm and 25 mm ID bearings required uneven spacers, so all bearings were standardised to 20 mm ID to reduce manufacturing error while maintaining sufficient strength.

Early CAD modelling showed that the SKF SY20TF bearings clashed with the extrusion frame. They were replaced with RS PRO UCPA204 short-base units, which fit the space but required underside mounting, incompatible with the frame. An L-bracket was considered but rejected due to insufficient stiffness. SKF P20 pillow block bearings were ultimately selected for their correct mounting orientation and compatibility with the frame.

The shaft layout evolved repeatedly as bearing choice dictated feasible placement. Initial layouts offered no practical bearing positions while maintaining gear meshing, and alternatives became unnecessarily complex. With the final bearings, Shaft 1 required vertical mounting to fit within the aluminium frame.

Taper-bushed sprockets were initially preferred for axial adjustability, but keys and endcaps were adopted instead, offering equivalent functionality at significantly lower cost.

4.3 Operations and Performance Verification

Based on the selected configuration, the values for input and output speed, torque, power, and actual modification have been updated in Table 10. The layout of each stage is shown in Figure 3.

Table 10: Input and output speed, torque, power, and the actual modification achieved.

	Speed [rpm]	Torque [Nm]	Basic Power [W]	Required Power [W]	Actual Modification
Input	2081.1	1.094		242.3	10.12
Output	205.7	10.316	222.2		

Performance verification consists of a bench test and a full-load track test. The bench test focuses on ensuring correct chain tension, sprocket alignment, and proper gear meshing. The full-load test is carried out on a straight track to verify the required top speed of 20–24 km/h and that the drivetrain can be easily attached and removed from the F24 car. Thermal rise and fastener integrity are assessed to ensure

durability. Safety requirements are followed throughout all testing procedures and a risk assessment for testing is provided in Appendix E. Product is verified using the criteria in Appendix D.

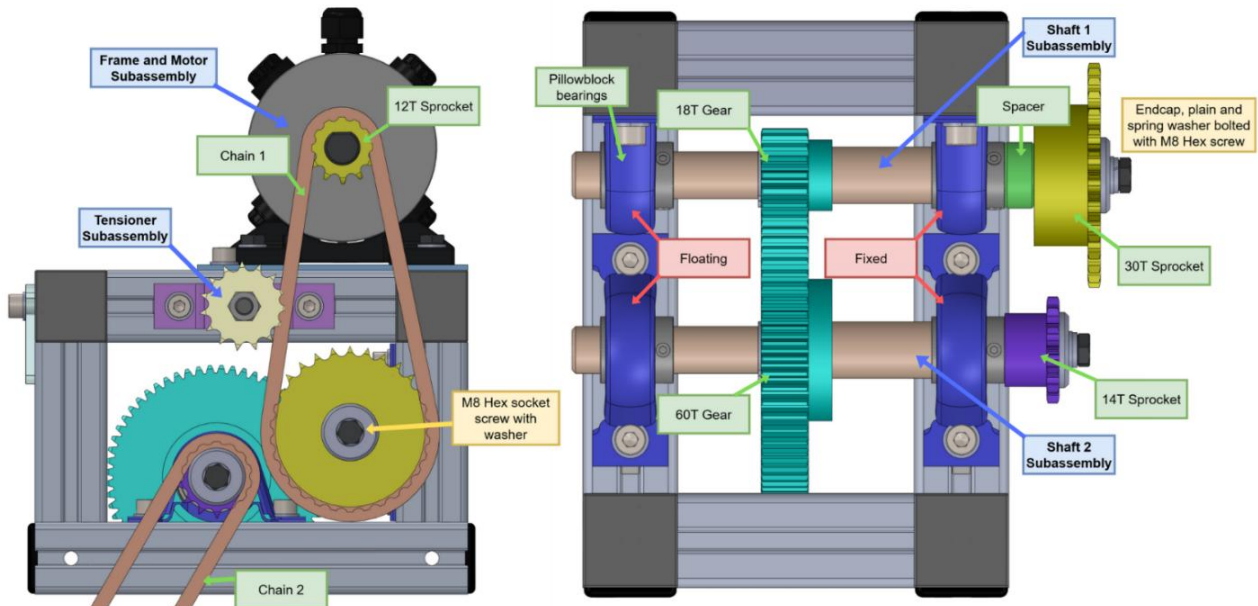


Figure 3: Diagram showing each stage of the drivetrain.

5. Manufacturing and Prototyping

5.1 Manufacturing Requirements, Constraints, and Processes

Manufacturing and assembling are performed in the Student Training Workshop (STW) during designated sessions. During these sessions, safety protocols are strictly followed. Stock materials are available at STW and items such as gears and sprockets are purchased from RS. Purchased parts are provided in Section 7.3.

Tolerance of machined parts follow the ISO standard of manufacturing. The tolerances chosen are slightly greater due to producer skills. Table 11 shows the tolerances for crucial components. For parts without specified tolerance values, it will follow the sheet tolerance value. Surface roughness is controlled between 0.4 to 1.6 and a total runout of 0.01 mm is set for shaft.

Table 11: Tolerance grade and value for each feature on a manufactured part.

Part	Feature	Tolerance Grade	Tolerance Value
Shaft	Main diameters	h6	-0.013 / 0.000 mm
Shaft	Keyway width	N9	-0.030 / 0.000 mm
Shaft	Keyway depth	-	0.000 / +0.100 mm
Shaft	Circlip groove diameter	-	-0.210 / 0.000 mm
Shaft	Circlip groove width	-	0.000 / +0.140 mm
Gears/Sprockets	Bore diameter	H7	0.000 / +0.021 mm
Gears/Sprockets	Keyway width	D10	+0.030 / +0.078 mm
Gears/Sprockets	Keyway depth	-	0.000 / +0.100 mm
Keys	Width	h9	-0.030 / 0.000 mm
Keys	Thickness	h11	-0.075 / 0.000 mm
Spacer	Bore diameter	H7	0.000 / +0.021 mm

Table 12 displays the machine used for every parts. Three manufacturing drawings of critical parts is shown in Appendix A. A decorative plate is designed to continue the “Frozen” movie franchise theme for

client purposes. After each part is manufactured, a quality check is performed, and a measurement report of the two most crucial components is created and can be seen in Appendix F.

Table 12: Material, manufacturing operation, and tools used for each part.

Part	Material	Manufacturing Operation	Machines/Tools Used
Shafts	Steel EN1A	Turning, boring, milling, tapping	CNC, lathe, mill, tap
Gears/Sprockets	Steel EN8	Turning, boring, broaching	Lathe, hydraulic press
Keys	Steel EN1A	Cutting, sanding	Hacksaw, sander
Spacer	Aluminium	Turning, boring	Lathe
Motor mount	Steel EN1A	Drilling, milling	Mill
Tensioner bracket	Aluminium	Drilling, milling	Mill
Decorative plate	Acrylic	Laser cutting	Laser cutter

5.2 Prototype Assembly

The prototype consists of four subassemblies as shown in Figure 4. The bill of material is in Appendix C and the drawing for the general assembly is shown in Appendix B. Table 13 shows the sequence of assembling each subassembly.

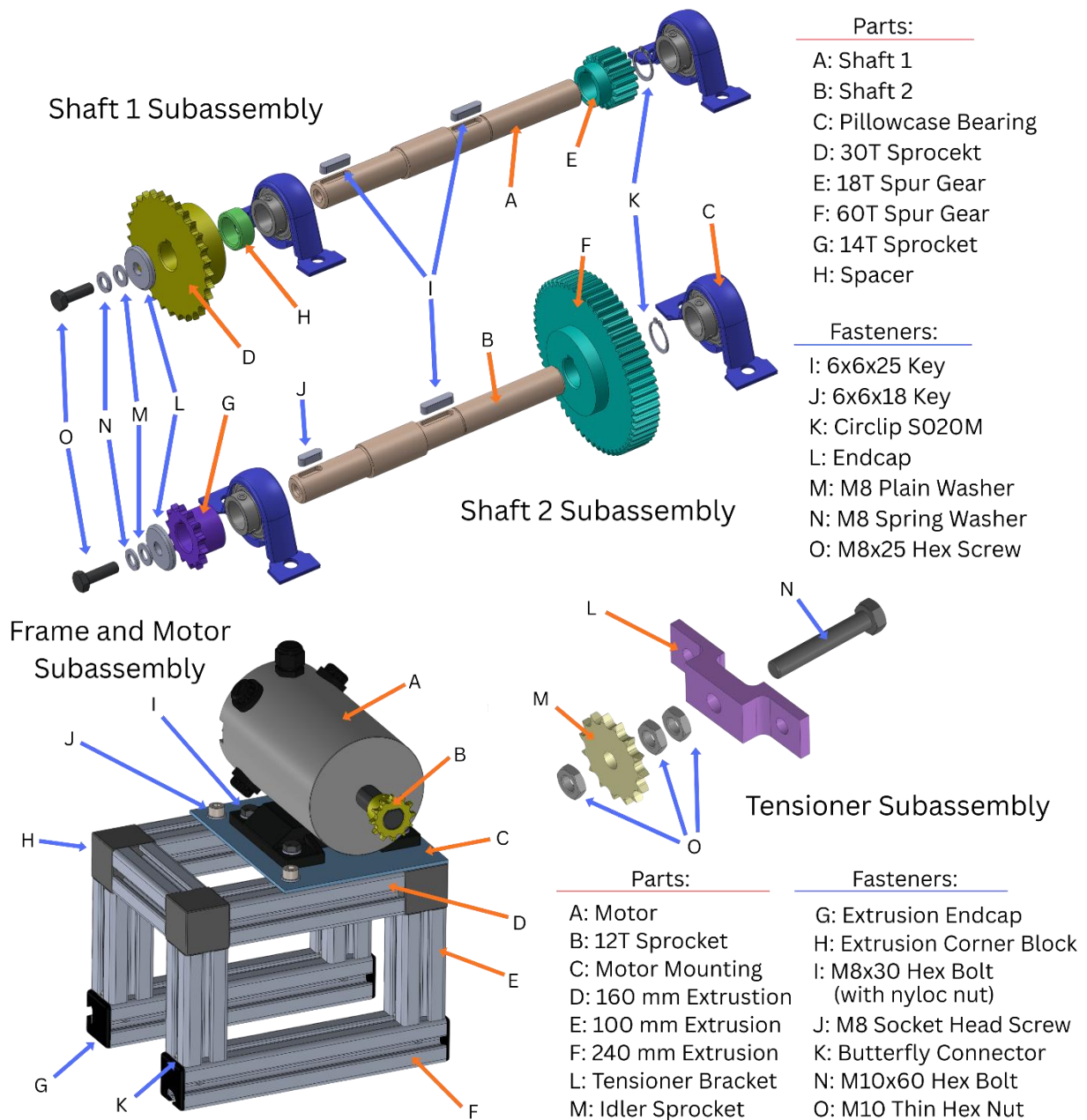
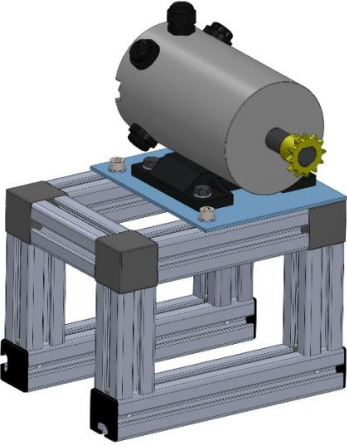
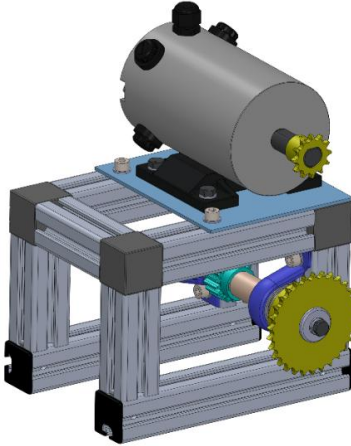
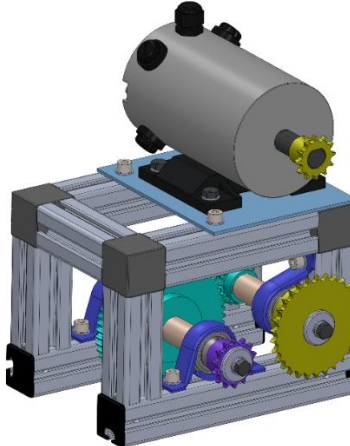
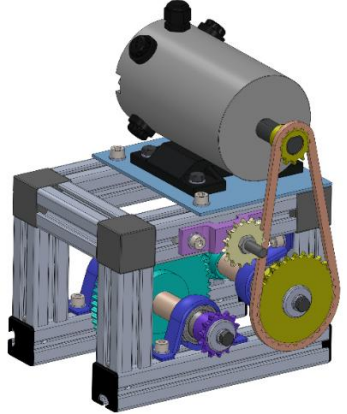
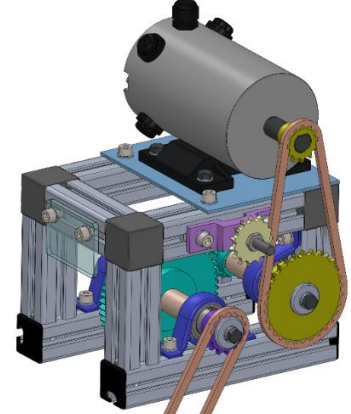
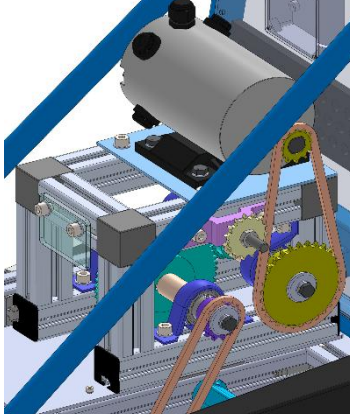


Figure 4: Exploded and annotated views for each subassembly.

Table 13: Assembly sequence for each subassembly.

 (1)	 (2)	 (3)
Assemble the frame and motor subassembly.	Secure shaft 1 subassembly to the side of the extrusion cassette.	Secure shaft 2 subassembly to the bottom extrusion with sprockets facing the same direction.
 (4)	 (5)	 (6)
Attach the 56 pitches chain onto the first stage sprockets and tension it using the tensioner assembly.	Secure the decorative plate onto the side of the cassette. Attach the 72 pitches chain onto the 14T sprocket.	Slide the module onto the rear of the F24 car. Connect the chain to the axle sprocket. Secure the cassette tightly and close cover.

All hex and socket head screws are tightened to 25 Nm and 30 Nm, respectively, and all screws uses a M8 washer. Screws are lubricated before tightening on shaft and bearings are greased.

The sprockets for both the first and last stages must also be aligned axially, with less than 0.5 mm of offset. To tension the first stage chain, a tensioner is placed on the aluminium extrusion. It could slide along the rail for flexible tensioning. For the final-stage chain connected to the axle, tension is set by repositioning the entire aluminium cassette until only a barely visible amount of slack remains.

6. Potential Failure Modes and Mitigation

Shaft yielding occurs when stresses exceed the material's yield strength, causing permanent deformation. This can result from stress concentrations due to sharp corners, keyways, or machining marks. Misaligned bearings can introduce excessive bending stresses, and poor lubrication may cause frictional heating and thermal stress. Chain derailment happens when the chain slips off the sprocket and

fails to engage properly. In this design, it may result from sprocket misalignment or incorrect chain tension, causing sagging and wear.

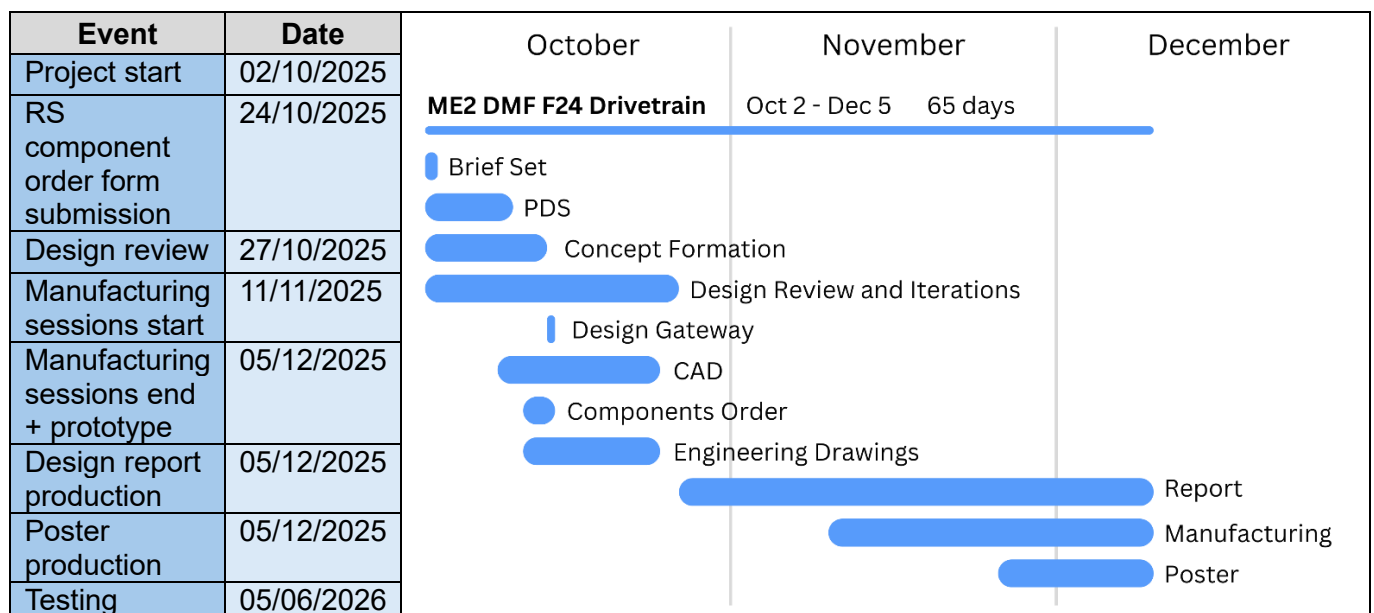
To reduce stress concentrations, generous radii are added at shoulders and transitions, sharp edges are avoided, and tolerances are controlled. Shaft diameters are selected to meet torque requirements with a safety factor of approximately 6. Slotted holes in the motor plate allow alignment adjustments during assembly. To prevent chain derailment, sprockets are aligned precisely during installation, using straight edges for accuracy. A tensioner is included for the chain between the 12T and 30T sprockets to prevent sagging. Chain tension between the 14T and 17T sprockets is set during final assembly.

7. Project Management

7.1 Project Timeline

The project spanned 10 weeks from initiation on 2nd October 2025 to testing on 5th June 2026. Concept generation starts immediately, and manufacturing only starts after receiving approval in the design gateway on the 27th October 2025. The prototype, design report, and poster are submitted on 5th December 2026. Table 14 summarises the key deadlines and displays the Gantt chart that outlines the workflow used to keep the team on schedule and ensure the prototype was ready for testing.

Table 14: Key deadlines and workflow timeline



7.2 Individual Time Log and Team Coordination

Figure 4 displays the number of hours spent by each team member. Note that the hours are rough estimates intended to illustrate the overall distribution of effort.

All documents including CAD files, engineering drawings, and spreadsheets were saved in a shared Google Drive folder. This allowed the team to work simultaneously on drawings and CAD. Meeting minutes and notes were taken in the logbook, with a designated chair and secretary each time. Team members communicate, update, and schedule meetings through WhatsApp.

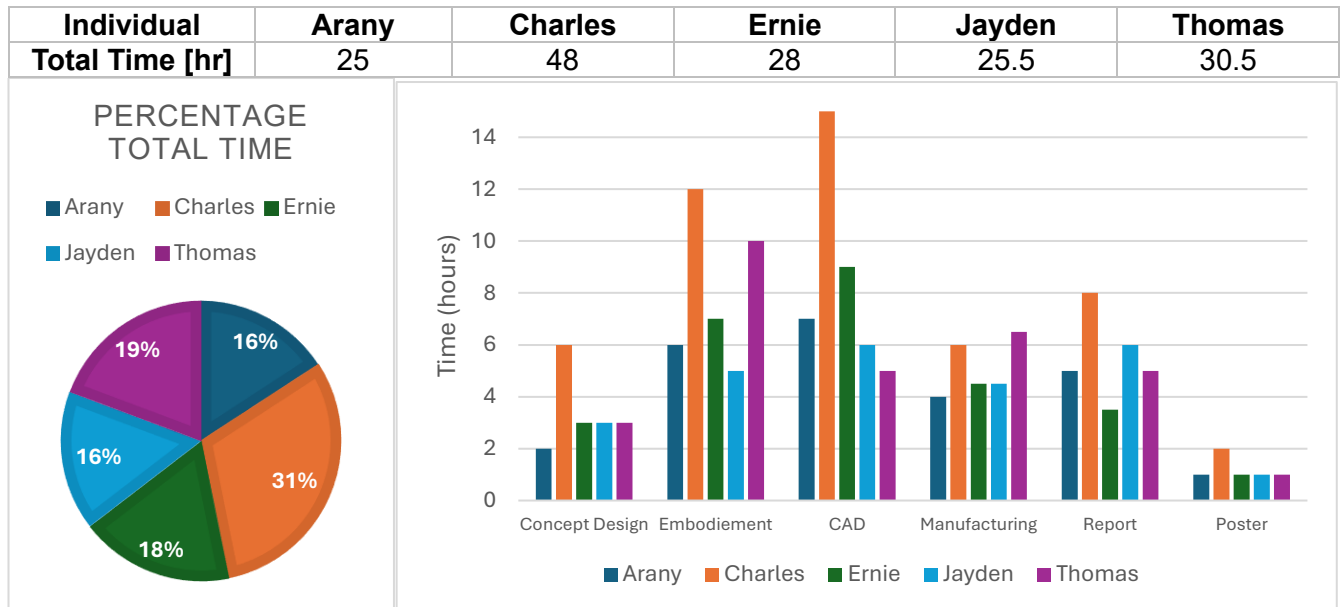


Figure 4: Individual time log and work break down.

7.3 Budget and Procurement

The project had a budget of £300 for purchasing parts from RS. Parts lists are handed to the Department for procurement. Table 15 shows the part list and the final budget.

Table 15: Purchase record showing the RS number, part description, quantity, and price.

RS No.	Part	Price (£)	Quantity	Total (£)
339-8726	Bearings	40.04	4	160.16
521-6682	Spur Gear M2 60T	91.47	1	91.47
521-6575	Spur Gear M2 18T	7.43	1	7.43
267-9067	Sprocket 06B-1 30T	7.4	1	7.4
267-9081	Sprocket 06B-1 14T	2.51	1	2.51
781-0766	Through Bore Sprocket (Tensioner)	30.05	1	30.05
Total				299.02

7.4 Project Risks and Mitigation

Table 16 summarises the key risks identified, and the control measures taken to minimise their effects.

Table 16: Project risks, its consequences and control measures.

Risk	Consequence	Control measure
Manufacturing did not finish on time.	Unfinished prototype.	Work was planned before hand to ensure efficiency and more time given.
Uneven team contribution.	Increased workloads on remaining members and lower quality outputs.	Regular meetings scheduled in person to ensure tasks are distributed evenly or according to each member's strength.
RS item delivery delays	Unfinished prototype.	Only in-stock or short lead-time components were ordered.
File and data loss.	Loss of essential design information.	Frequent backups on cloud and local drive; Crucial design decisions on paper.
Facility shutdowns and strikes.	Unfinished prototype.	Manufacturing plans created in advance; Critical parts were prioritised early.

8. Conclusion

The aim of this project was to design a transmission for the F24 car that reduces the maximum speed of the vehicle to 20-24 km/h, which could be assembled with a budget of £300 using a combination of purchased parts from RS Components and parts manufactured on the machines available in the STW. One of the toughest challenges faced in the process was fitting the design within the aluminium frame. Additionally, the sprockets provided with the car's motor and axle heavily influenced the transmission design. After several iterations, the final solution was a three-stage transmission involving two simplex 06B chain stages and one spur gear stage.

According to calculations made during the transmission design, the car is expected to operate at 20 km/h. Further calculations of the forces on the components of the transmission under regular operation were carried out. Given that shaft two will experience higher radial and axial loads, only the safety factor for this shaft was calculated, and it was found to be 5.8. The 18T and 60T gears had safety factors for tooth bending of 8.16 and 11.43, and safety factors for contact stress of 2.12 and 2.27, respectively. All bearings were found to experience loads below their dynamic load rating. Whilst the limited space for the design to fit caused several problems during the design process, the final solution can be mounted on the car, and the safety cover can still close. With all the above factors considered and the transmission costing £299.02, the requirements for the transmission were met.

At this moment in time, the prototype is not fully complete. One of the bearings delivered does not match what was listed on the order form and does not fit the design. This causes shaft 1 to be missing one bearing; however, when this part is received, the prototype can be finished entirely.

In the future, it is essential to make the design as low-maintenance as possible and ensure the transmission runs smoothly for extended periods. A step towards this would be to design a separate cover for the gear stage. Although the safety cover primarily protects it, some gaps could allow dust and debris to enter, leading to accelerated gear wear.

Bench tests will be performed after assembly is complete and the load testing session will involve driving the car in a straight line. Whilst this will confirm whether the design reduces the car's speed as intended, it will not assess the vehicle's cornering ability. Currently, the motor is mounted on top of the aluminium box, which raises the car's centre of mass and reduces cornering ability. Future work might be to rearrange the layout of the transmission so that the motor can be situated as close to the ground as possible, therefore making the car more suitable for use in a race situation.

9. Acknowledgements and References

The success and outcome of this project required a lot of guidance and assistance. We would like to thank the Student Teaching Workshop staff, our personal tutor, and the Department of Mechanical Engineering at Imperial College London for their technical guidance, material support, and assistance with manufacturing.

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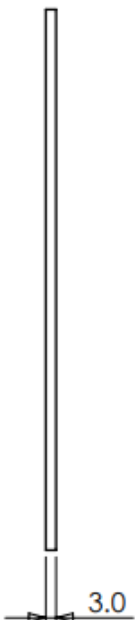
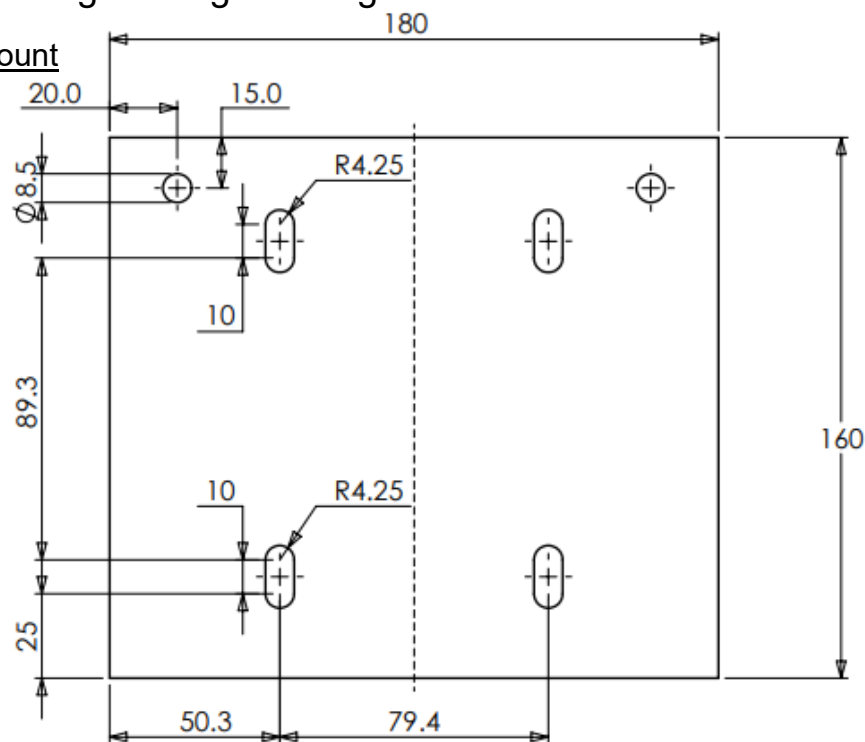
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10. Appendices

Appendix A: Engineering Drawings

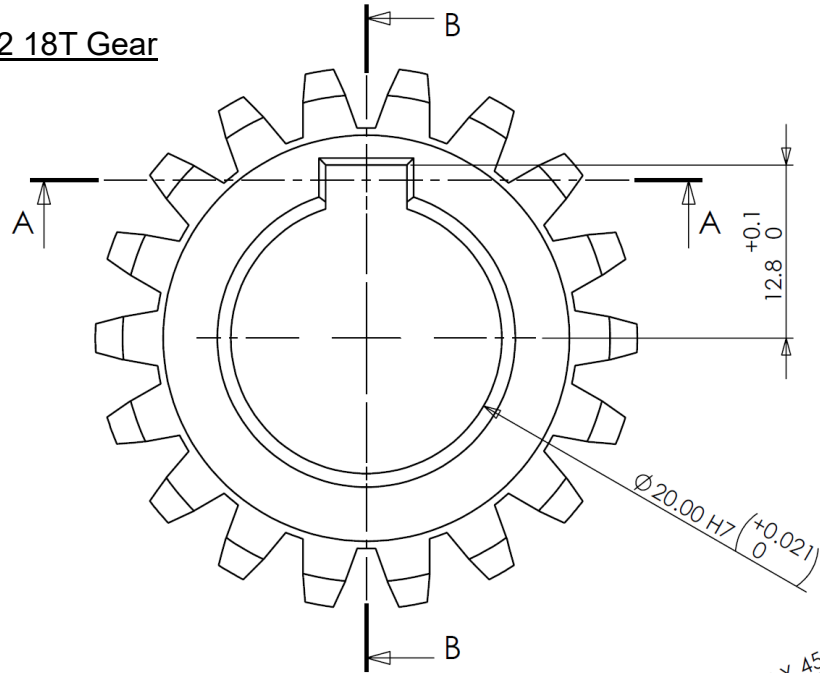
A.1 Motor Mount



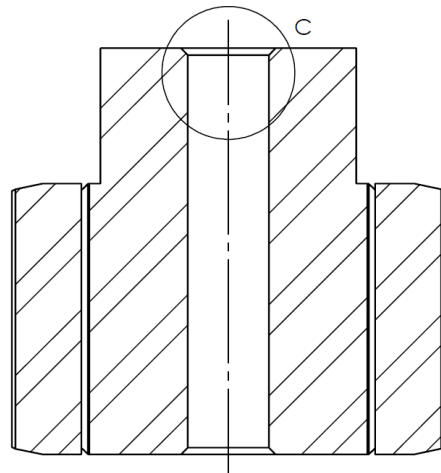
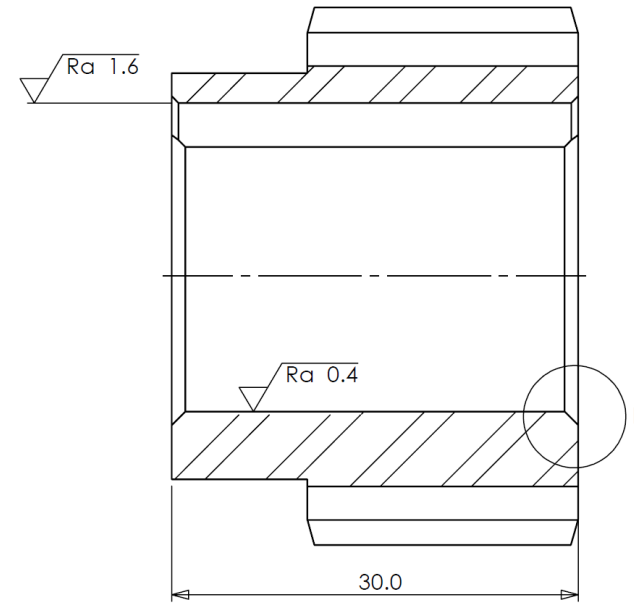
GENERAL NOTES:
DEBUR ALL SHARP EDGES
8.5MM HOLES ARE CLEARANCE FOR M8 BOLTS,
Design is symmetric about the centreline

TOLERANCES		THIRD ANGLE PROJECTION	MATERIAL: MILD STEEL EN1A	TITLE: MOTOR MOUNT	Imperial College London Department of Mechanical Engineering
X = ± 0.5	ANGULAR ± 1°				
X.X = ± 0.1	SURFACE FINISH		ALL DIMENSIONS ARE IN MILLIMETRES	DWG No. DMF2_P21_P13	SHEET 1 OF 1
X.XX = ± 0.02	MACHINED FACES Ra 6.3				
	NAME	DATE	DO NOT SCALE DRAWING	A4 SCALE 1:2	REVISION
DRAWN	T. SUTHERLAND	06/11/25			
CHECKED	E. COLLINSON	06/11/25			
APPROVED	E. COLLINSON	06/11/25			

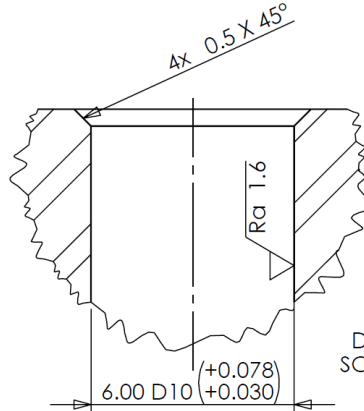
A.2 18T Gear



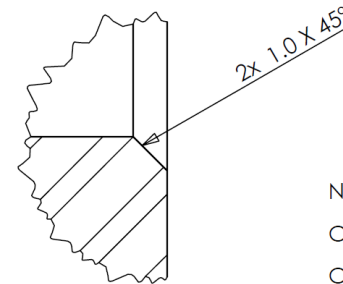
SECTION B-B



SECTION A-A



DETAIL C
SCALE 5 : 1



DETAIL D
SCALE 5 : 1

NOTE:

CHAMFER 0.5mm ON KEYWAY EDGES

CHAMFER 1.0 mm ON BORE EDGES

TOLERANCES		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London Department of Mechanical Engineering		
X	=± 0.5	ANGULAR ±1°		STEEL EN8		18T SPUR GEAR				
X.X	=± 0.1	SURFACE FINISH								
X.XX	=± 0.02	MACHINED FACES Ra 6.3								
		NAME		DATE		ALL DIMENSIONS ARE IN MILLIMETRES				
DRAWN	J. LEE		21.10.2025		DO NOT SCALE DRAWING		DWG No.			
CHECKED	C. HUNG		22.10.2025				DMF2_P21_P04			
APPROVED	E. COLLINSON		24.10.2025		A4		SCALE 2:1		SHEET 1 OF 1	
								REVISION A		

**Imperial College
London**

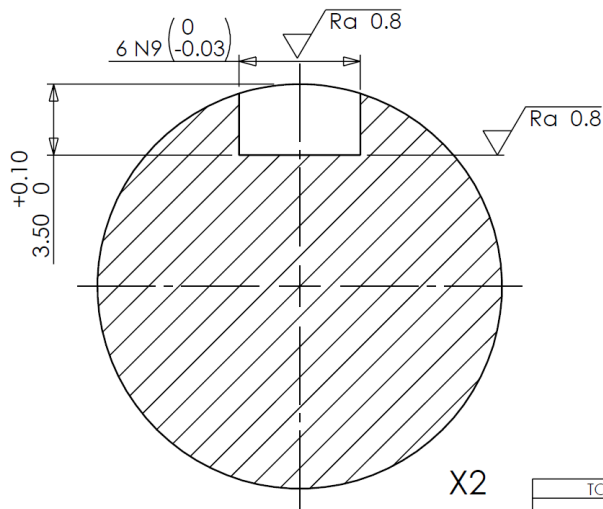
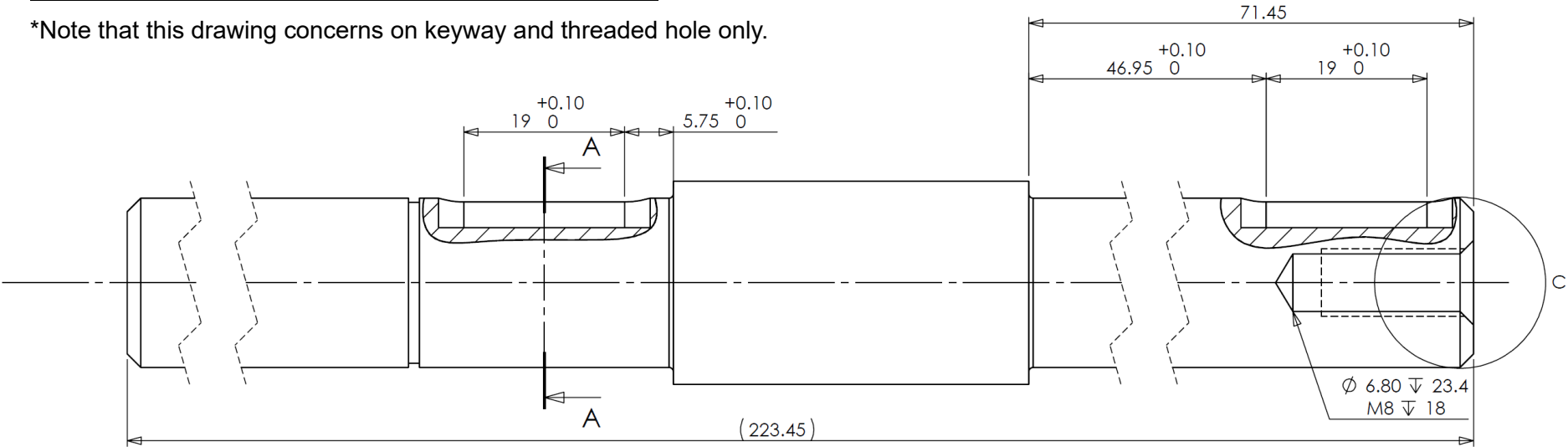
**Department of
Mechanical Engineering**

SHEET 1 OF 1

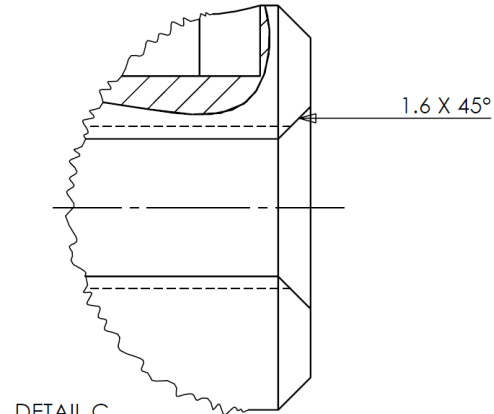
REVISION A

A.3 Shaft 1 Drawing for Keyways and Threaded Hole

*Note that this drawing concerns on keyway and threaded hole only.



SECTION A-A
SCALE 3 : 1

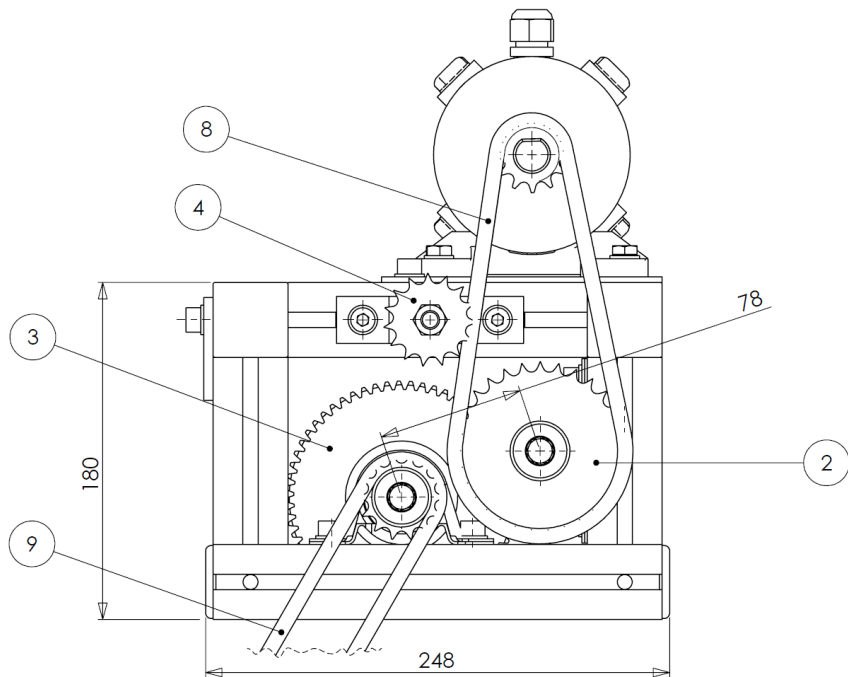
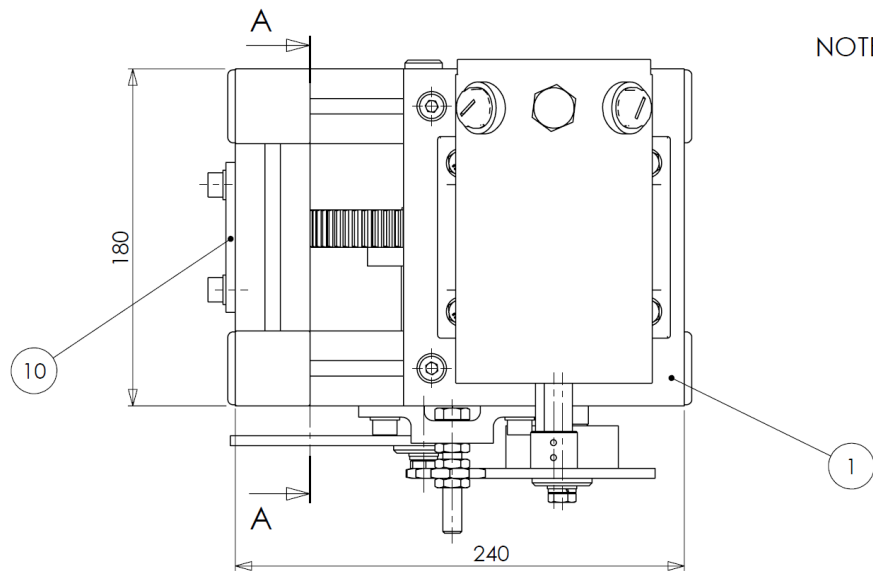


DETAIL C
SCALE 3 : 1

- NOTES
- 1. CNC SHAFT DIAMETERS
 - 2. MILL KEYWAY
 - 2. TURN AND BORE CIRCLIP GROOVE AND BOLT HOLE
 - 4. MANUAL TAP THREAD

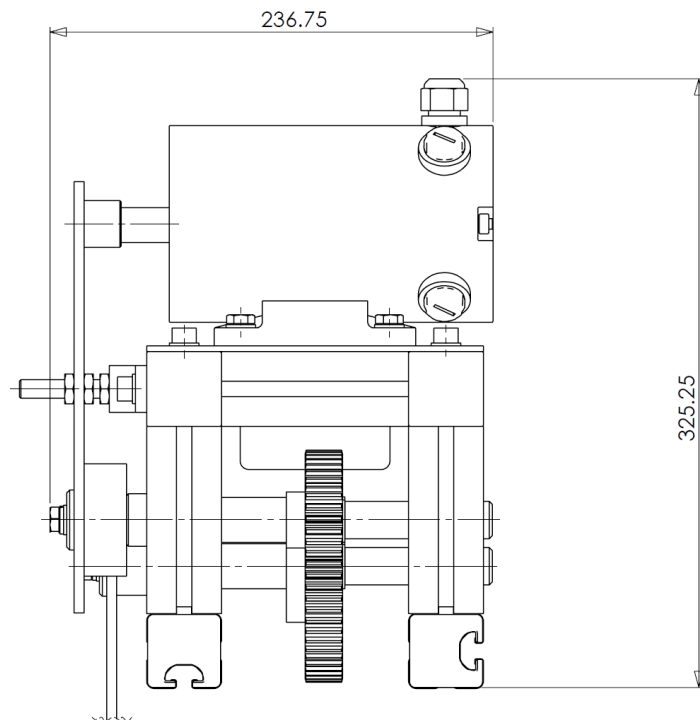
TOLERANCES		THIRD ANGLE PROJECTION		MATERIAL: STEEL EN1A		TITLE: SHAFT 1		Imperial College London Department of Mechanical Engineering	
X	± 0.5	ANGULAR	±1°						
X.X	± 0.1	SURFACE FINISH	MACHINED	ALL DIMENSIONS ARE IN MILLIMETRES					
X.XX	± 0.02	FACES	Ra 6.3						
NAME		DATE		DO NOT SCALE DRAWING		DWG No. DMF2_P21_P01			
DRAWN	C. HUNG	21.10.2025							
CHECKED	T. SUTHERLAND	24.10.2025		A4	SCALE 3:2	SHEET 1 OF 1		REVISION B	
APPROVED	E. COLLINSON	30.10.2025							

Appendix B: General Assembly



NOTES

1. ASSEMBLE EACH SUBASSEMBLIES FIRST
2. INSTALL SHAFT 1 ASSEMBLY BEFORE SHAFT 2 ASSEMBLY
3. INSTALL CHAIN ON SPROCKETS BEFORE SECURING BEARINGS
4. SLIDE TENSIONER TILL FIRST STAGE CHAIN IS APPROPRIATELY TENSIONED
5. SECURE EACH SCREW IN PLACE BEFORE TIGHTENING
6. ENSURE ACCURATE CENTRE DISTANCE BETWEEN SHAFTS
7. SLIDE ENTIRE CASSETTE TO TENSION LAST STAGE CHAIN



TOLERANCES			MATERIAL:	TITLE:	Imperial College London Department of Mechanical Engineering	
X = ± 0.5	ANGULAR ±1°					
X.X = ± 0.1	SURFACE FINISH					
X.XX = ± 0.02	MACHINED FACES Ra 6.3					
NAME		DATE	DO NOT SCALE DRAWING			DWG No. DMF2_P21_A01
DRAWN	C. HUNG	23.10.2025	A3 SCALE 1:2.5			
CHECKED	T. SUTHERLAND	24.10.2025				
APPROVED	C. HUNG	16.11.2025				
			SHEET 1 OF 2		REVISION B	

Appendix C: Bill of Materials

Table 17 displays all components used for the drivetrain including the material the RS code.

Table 17: Bill of Materials

Part Number	Description	Quantity	Material	Notes
DMF2_P21_P01	SHAFT 1	1	Steel EN1A	
DMF2_P21_P02	SHAFT 2	1	Steel EN1A	
DMF2_P21_P03	PILLOW BLOCK BALL BEARING 20 ID	4		RS 339-8726
DMF2_P21_P04	SPUR GEAR 18T	1	Steel EN8	RS 521-6575
DMF2_P21_P05	SPUR GEAR 60T	1	Steel EN8	RS 521-6682
DMF2_P21_P06	SPROCKET 06B-1 14T	1	Steel EN8	RS 267-9081
DMF2_P21_P07	SPROCKET 06B-1 30T	1	Steel EN8	RS 267-9067
DMF2_P21_P08	KEY 6X6X18	1	Steel EN1A	
DMF2_P21_P09	KEY 6X6X25	3	Steel EN1A	
DMF2_P21_P10	CIRCLIP S020M	2		
DMF2_P21_P11	SPACER 20 ID	1	Aluminium T6	
DMF2_P21_P12	ENDCAP 9 ID	2		
DMF2_P21_P13	MOTOR MOUNTING PLATE	1	Steel EN1A	
DMF2_P21_P14	PLAIN WASHER M8	22		
DMF2_P21_P15	SPRING WASHER M8	2		
DMF2_P21_P16	T-SLOT NUT M8	16		
DMF2_P21_P17	HEX SCREW M8X25	4		
DMF2_P21_P18	HEX SOCKET HEAD SCREW M8X16	14		
DMF2_P21_P19	CHAIN 06B-1 56 PITCHES	1		
DMF2_P21_P20	CHAIN 06B-1 72 PITCHES	1		
DMF2_P21_P21	TENSIONER BRACKET	1	Aluminium T6	
DMF2_P21_P22	TENSIONER SPROCKET	1		
DMF2_P21_P23	HEX BOLT M10X60	1		
DMF2_P21_P24	THIN HEX NUT M10	3		
DMF2_P21_K01	DECORATIVE PLATE	1	Acrylic	
GP1106	SPROCKET 12T 06B-1	1		
GP1227	NYLOC NUT M8	2		
GP1237	HEX BOLT M8x30	2		
GP1248	ELECTRIC MOTOR WITH MOUNT	1		
FYP005	CONNECTING CLIP 10200-08	4		
FYP 201	CORNER CONNECTOR 40x40	4		
FYP202	ALUMINIUM EXTRUSION 40x40x240	2		
FYP203	ALUMINIUM EXTRUSION 40x40x100	6		
FYP204	ALUMINIUM EXTRUSION 40x40x160	2		
FYP205	SLOTTED END CAP 40x40	4		

Appendix D: Test Criteria

Table 18 is the criteria the product will assessed on during testing.

Table 18: Assessment criteria for full-load test.

Parameter	Verification Method	Pass/Fail Criteria
Chain Tension Stability	Visual inspection and/or tension gauge.	No derailment or slack beyond tolerance.
Gear Engagement Quality	Inspect full tooth contact and no skipping.	No visible skipping or wear.
Thermal Rise	IR thermometer.	No overheating of bearings or gears above 60°C
Fastener Integrity	Torque wrench post-test.	Specific torque requirements: hex bolts 25 Nm and socket head screws 30 Nm.
Output Shaft Speed	Tachometer and timed straight track test.	Matches theoretical RPM and speed within margin.

Appendix E: Risk Assessment

Table 19 shows the risk assessment for testing the prototype and the control measures for mitigation.

Table 19: Risk assessment and control measures on prototype testing.

Hazards	Who it can harm	How it can harm	Control Measures
Chain Derailment	Operator, Bystanders	Whip injury, sudden jamming causing hand traps or impact	Correct chain tension and alignment, keep hands clear, emergency stop available.
Rotating sprockets, gears and chains	Operator, Bystanders	Entanglement, lacerations from exposed teeth	Close cover, keep hands clear, emergency stop available.
Pinch points at meshing gears	Operator	Fingers trapping and crush injuries	No hands near live mechanisms.
Ejected parts from failure	Operator, Bystanders	Impact injuries to eyes/face	Face shield and safety glasses.
Sharp edges on sprockets/gears, mostly from STW	Operator	Cuts during handling and fitting	Deburr components, cut-resistant gloves.
Hot components after prolonged testing	Operator	Contact burns from friction-heated parts	Allow cool down, temperature checks.
Loose clothing, hair, jewellery	Operator	Entanglement in rotating parts	Tie hair back, remove jewellery, wear fitted clothing and sleeves

Appendix F: Measurement Report

Table 20 presents the measurement report for Shaft 1 and the 18T Spur Gear dimensions.

Note that the Shaft 1 body is manufactured using CNC machining; therefore, its diameter and length dimensions are provided separately in the Shaft 1 CNC drawing (DMF_P21_K02).

Table 20: Measurement report on Shaft 1 and 18T Spur Gear.

Part Number	DMF2_P21_P01	Description	Shaft 1	Material	Steel EN1A	
No	Part Detail		Required Dimensions	Lower Bound	Upper Bound	Actual Dimensions
1	Shaft Length (R)		71.45	71.43	71.47	71.47
2	Keyway (R) from Shoulder		46.95	46.95	47.05	46.95
3	Keyway (R) Length		19.0	19.0	19.1	19.1
4	Keyway (R) Width		6.00	5.97	6.00	5.97
5	Keyway (R) Depth		3.5	3.5	3.6	3.6
6	Keyway (L) from Shoulder		5.75	5.75	5.85	5.75
7	Keyway (L) Length		19.0	19.0	19.1	18.9
8	Keyway (L) Width		6.00	5.97	6.00	6.02
9	Keyway (L) Depth		3.50	3.50	3.60	3.49
10	Threaded Hole Diameter		6.80	6.78	6.82	6.77
11	Threaded Hole Depth		23.4	23.3	23.5	23.49
12	Chamfer (Hole) x 45°		1.6	1.5	1.7	1.51

Part Number	DMF2_P21_P04	Description	18T Spur Gear	Material	Steel EN8	
No	Part Detail		Required Dimensions	Lower Bound	Upper Bound	Actual Dimensions
1	Thickness		30.0	29.9	30.1	30.08
2	Bore Diameter		20.000	20.000	20.021	20.070
3	Keyway Width		6.000	6.030	6.078	6.08
4	Keyway Depth from Centre		12.8	12.8	12.9	12.9
5	Chamfer (Bore) x 45°		1.0	0.9	1.1	1.15
6	Chamfer (Keyway) x 45°		0.5	0.4	0.6	0.35

Appendix G: Speed, Torque, and Power Calculations

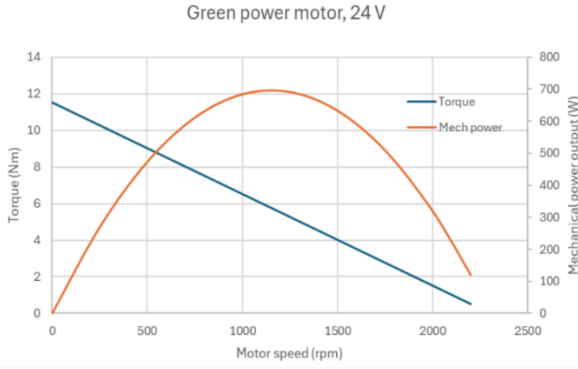


Figure 5: DC motor characteristics. (Mohammed. 2025)

Figure 5 displays the torque and power characteristic of the electric motor. The torque line is described in Equation 6 where T_m is the motor torque in Nm and z_{in} is the motor speed in rpm. By using the relationship between stages, Equation 8 could be solved to find modification G. T_{req} is 10.3 Nm from frictional and drag forces, η is the efficiency at 0.93, and z_{out} is 208.9 rpm.

$$T_m = (-5 \times 10^{-3})z_{in} + 11.5 \quad (6)$$

$$\eta GT_m = T_{req} \Rightarrow T_m = \frac{T_{req}}{\eta G}, z_{in} = Gz_{out} \Rightarrow \frac{10.3}{0.93G} = (-5 \times 10^{-3}) * 208.9G + 11.5$$

$$G = 9.94$$

Appendix H: Gear Stress Example Calculation

Table 21 shows the key properties of the pinion (Ihracska et al., 2023). Pitch line velocity and the dynamic factor is found as follows. Using the Lewis Equation, the minimum face width is found and is smaller than the pinion face width as shown. The safety factor for the pinion due to tooth bending is determined to be 8.16.

Table 21: Key properties of the pinion.

Pitch Diameter [mm]	Speed [rpm]	Material	Permissible Stress [MPa]	Pressure Angle [°]	Transmitted Load [N]	Lewis Factor
36	832.4	EN8	131	20	149.7	0.29327

$$v = \frac{36 \times 10^{-3}}{2} * 832.4 * \frac{2\pi}{60} = 1.569 \text{ ms}^{-1}, K_v = \frac{6.1}{6.1 + 1.569} = 0.795$$

$$w = \frac{149.7}{0.795 * 2 * 0.29327 * 131 \times 10^6} = 2.45 \text{ mm} < 20 \text{ mm} \therefore \text{valid}$$

To find the safety factor due to contact stress, Equation 2 and 3 is used. Table 22 shows the key values of the pinion from AGMA charts and previous calculations. The elastic coefficient C_p and geometry factor is calculated as shown. ϕ_t is the pressure angle, and G is the modification.

Table 22: Key constants and values from AGMA charts and previous calculations.

Poisson's Ratio, ν	Overload Factor, K_0	Load Distribution Factor, K_m
0.3 (The Engineering Toolbox, 2007)	1.25	1.20
Elastic Modulus, E [GPa]	Dynamic Factor, K_v	Size Factor, K_s
207 (The Engineering Toolbox, 2003)	0.795	1
Surface Condition Factor, C_f	Temperature Factor, K_T	Reliability Factor, K_R
1	1	0.85
Allowable Contact Stress, S_c [MPa]	Stress Cycle Factor, Z_n	Hardness Ratio Factor, C_H
644	0.755	1

$$C_p = \sqrt{\frac{1}{\pi \left(\frac{1 - \nu_p^2}{E_p} + \frac{1 - \nu_g^2}{E_g} \right)}} = \sqrt{\frac{1}{\pi \left(\frac{1 - (0.3)^2}{207 \times 10^9} + \frac{1 - (0.3)^2}{207 \times 10^9} \right)}} = 190272 \text{ Pa}^{0.5}$$

$$I = \frac{\cos(\phi_t) \sin(\phi_t)}{2} * \frac{G}{G + 1} = \frac{\cos(20) \sin(20)}{2} * \frac{60/18}{60/18 + 1} = 0.123$$

The pinion stress cycle factor Z_n , and the allowable bending stress S_C is calculated as follows, where N is the number of cycles, L is the service life in hours, and H_B is the Brinell Hardness given as 200.

$$N = 60 * n * L = 6 * 832.4 * 30000 = 1.49832 \times 10^9$$

$$Z_n = 2.466N^{-0.056} = 2.466(1.49832 \times 10^9)^{-0.056} = 0.755$$

$$S_C = 2.22H_B + 200 = 2.22(200) + 200 = 644 \text{ MPa}$$

The absolute contact pressure and safety factor could be calculated as shown.

$$\sigma_c = C_p \sqrt{FK_0 K_V K_S \frac{K_m C_f}{wd I}} = 190272 * \sqrt{149.7 * 1.25 * 0.795 * 1 * \frac{1.20}{(20 * 36) \times 10^{-6}} \frac{1}{0.123}} = 269.5 \text{ MPa}$$

$$S_H = \frac{S_C Z_n C_H}{\sigma_c K_T K_R} = \frac{644}{269.5} * \frac{0.755 * 1}{1 * 0.85} = 2.12$$

Appendix I: Bearing Life Example Calculation

Starting off by considering the forces axially and radially. F_a (axial force) = 0 for spur gears, sprockets and chains, so we only consider radial load. Force acting on spur gear was calculated using Equation 7 and 8. T is the torque and ϕ is the pressure angle of the gear tooth.

$$F_t \text{ (tangential force, Y plane)} = \frac{T}{r} \quad (7)$$

$$F_r \text{ (radial force, X plane)} = F_t \tan(\phi) \quad (8)$$

Force acting on the chain/sprocket was calculated using the Equation 9. T is the torque, k is a constant representing the slack-to-tight tension ratio of the chain, and r_s is the radius of the sprocket. Then, by taking moments about the bearings, the force was calculated for both the x and y direction for every bearing. Then to calculate the resultant force (F_R), Equation 10 was used.

$$F_{sprocket} \text{ (Y plane)} = \frac{T(1+k)}{(1-k)r_s} \quad (9), \quad F_R = \sqrt{F_x^2 + F_y^2} \quad (10)$$

C_{min} is the minimum dynamic load rating, L_{10} is the bearing life in $\times 10^6$ revolutions, F is the radial load and k is a constant ($k = 3$ for ball bearings). Given $n = 832.4$ rad/s and $F = 124.67$ N for bearing A, L_{10} is calculated and substituted into Equation 5. C_{min} is compared against the dynamic load rating.

$$30000 \times 832.4 \times 60 = 1500 \times 10^6 \text{ revs} \Rightarrow C_{min} = 124.67(1500)^{\frac{1}{3}} = 1427.12 < 12700 \therefore \text{valid}$$

Appendix J: Stress Analysis for Shaft 2

All forces acting on the shafts, including reaction, weight, and tangential forces. are depicted in a Free Body Diagram. The tangential forces are calculated using Equation 7 and 9. The reaction forces in x and y direction for each bearing are determined by equating the sum of forces and moments to zero.

A Bending Moment Diagram and Shear Force Diagram is drawn for the shafts. The maximum bending and shear stress due to torsion for each segment is determined using Equation 11 and 12.

$$\sigma = \frac{My}{I} \quad (11), \quad \tau = \frac{Tr}{J} \quad (12)$$

The maximum bending stress is 38.3 MPa and the maximum shear stress is 8MPa. These are lower than the maximum allowable bending stress and shear stress of 145 MPa and 72.5MPa respectively.